

PreservaSense: Monitoring and Mitigating Risk of Intraoperative Pressure Injuries (IPI)

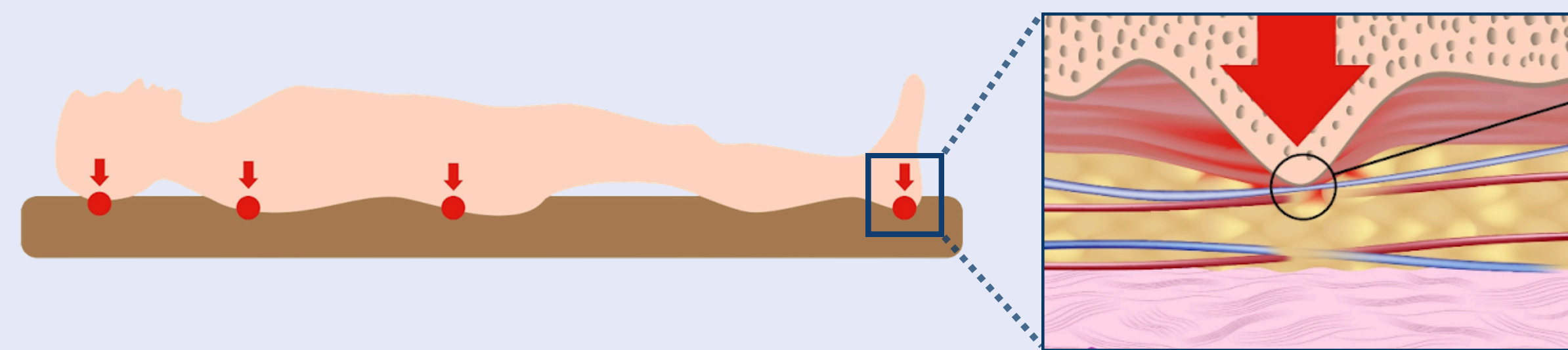


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Clinical Background

- Intraoperative Pressure Injuries (IPI):** when mechanical loading over **bony prominences** obstructs capillary flow, causing **ischemia** and eventual **tissue death**



- 8.5%** of all surgeries greater than **3 hours** result in IPIs¹

2.5 million annual cases
Treatment costs range **\$20,900 - \$151,700**

Issues with Current Procedure



No quantitative tool is implemented to monitor risks of IPIs



Health factors (e.g., obesity, malnutrition, age), surgery duration, and patient positioning lead to specific risk profiles

Our Mission

Our mission is to develop a method to **non-invasively monitor and mitigate risk of early-stage pressure injuries** for patients undergoing **surgeries greater than 3 hours**, to **reduce pressure complications**, thereby **shortening hospital stay and cost**, and **improving patient outcomes**.

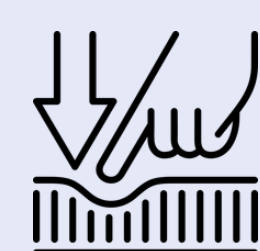
Design Specifications



Pressure Range
0-120 mmHg



Application Time per patch
< 1 min



Max Hardness
Shore 00-70



Vibration Frequency
~47 - 53 Hz

Cost Analysis

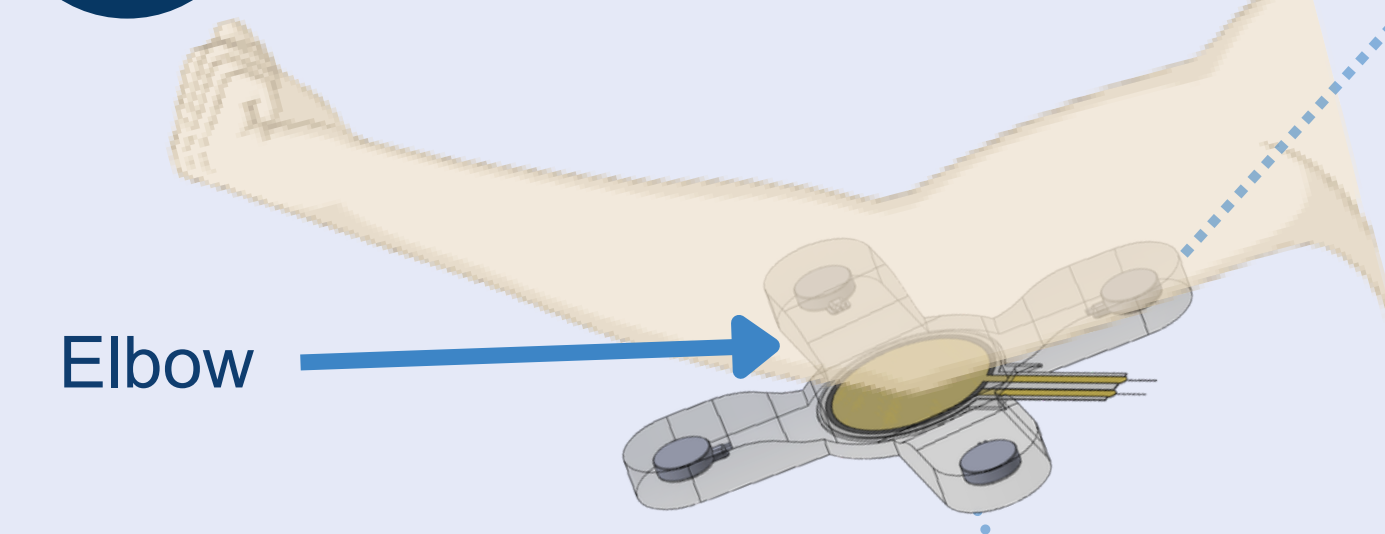
Value-Based Purchasing:
Hospitals are responsible for paying patients' hospital-acquired IPI (HAPI)



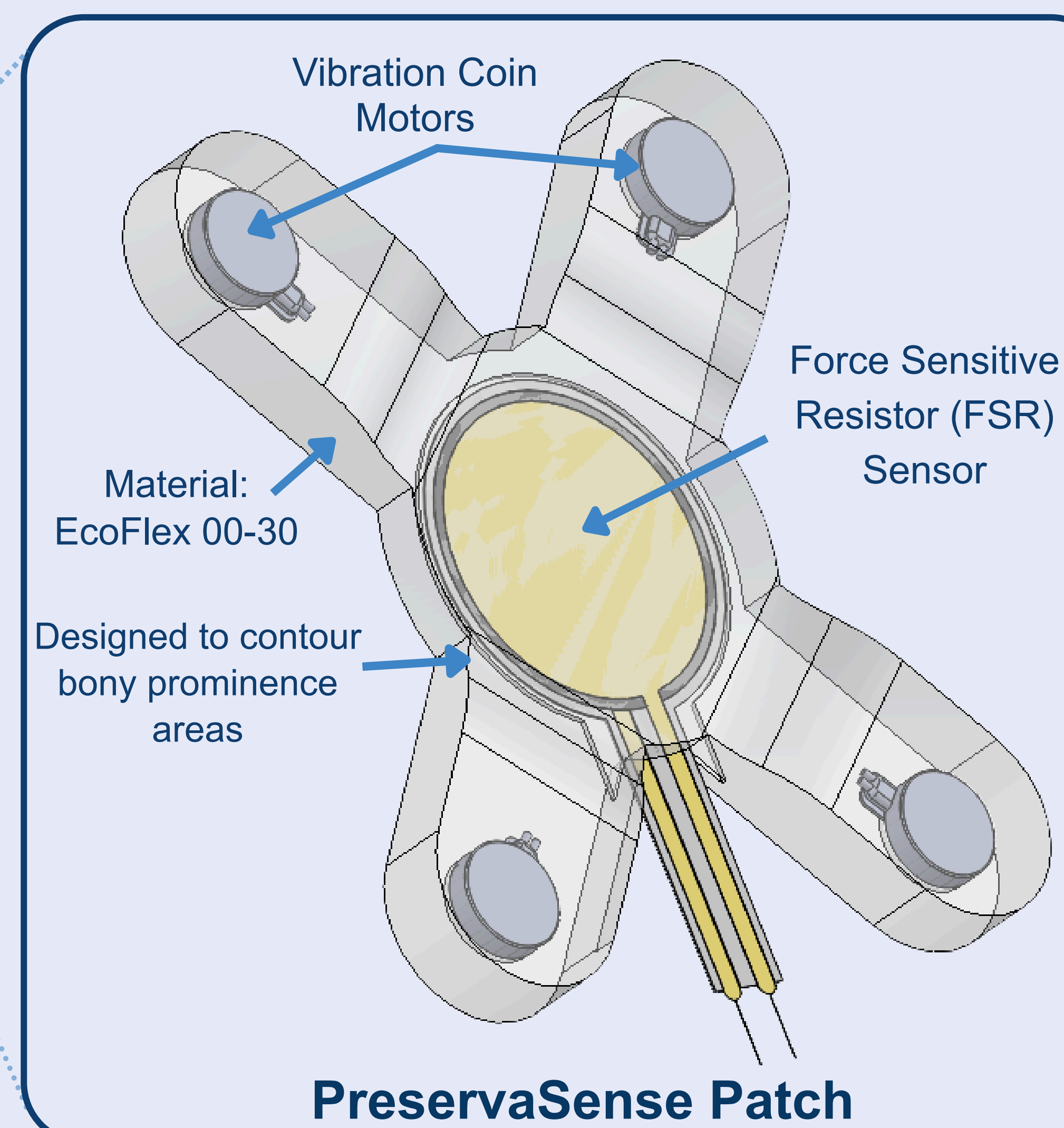
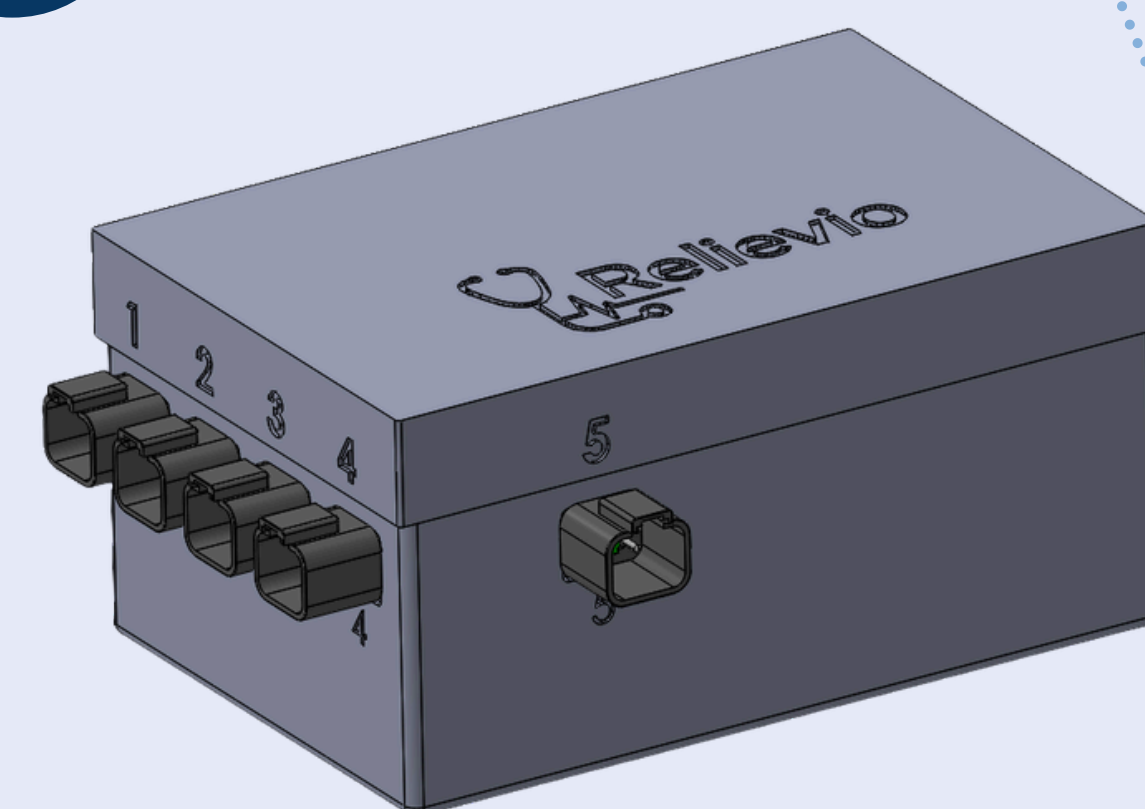
Reduce up to 80% of the costs associated with treating HAPI

Our Solution: PreservaSense

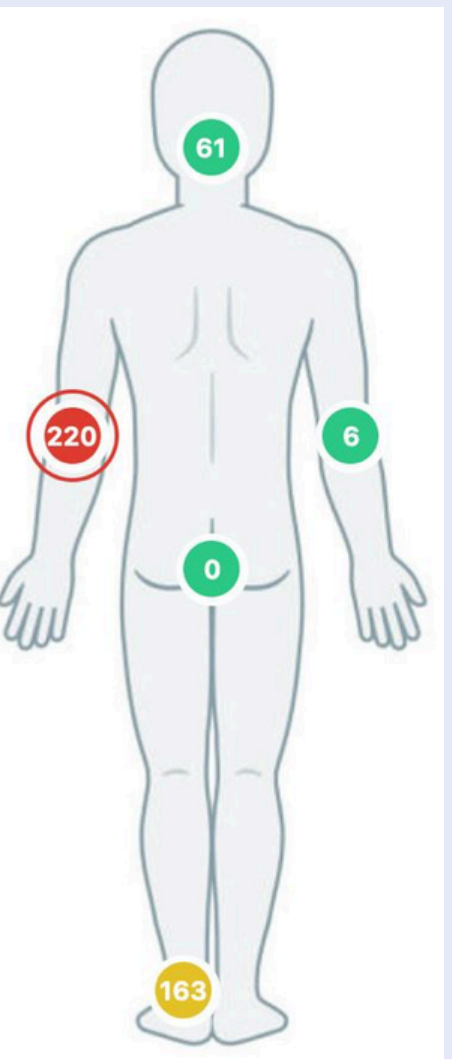
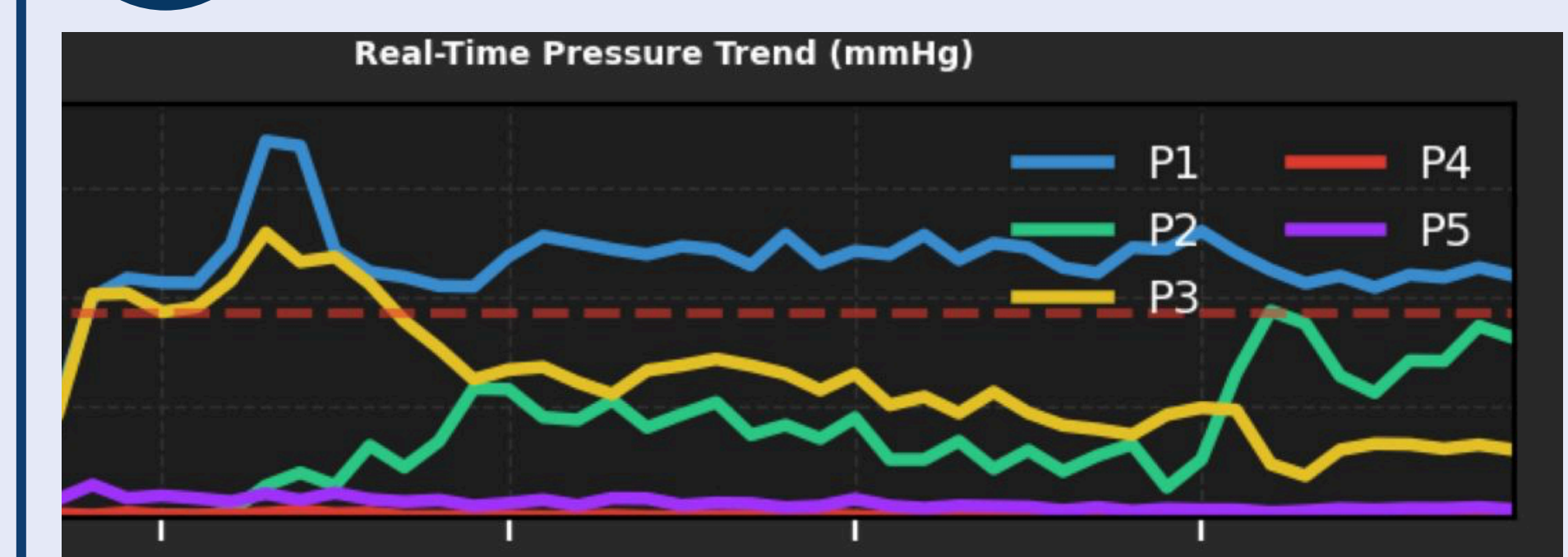
- 1 Wrap PreservaSense around the bony prominence area



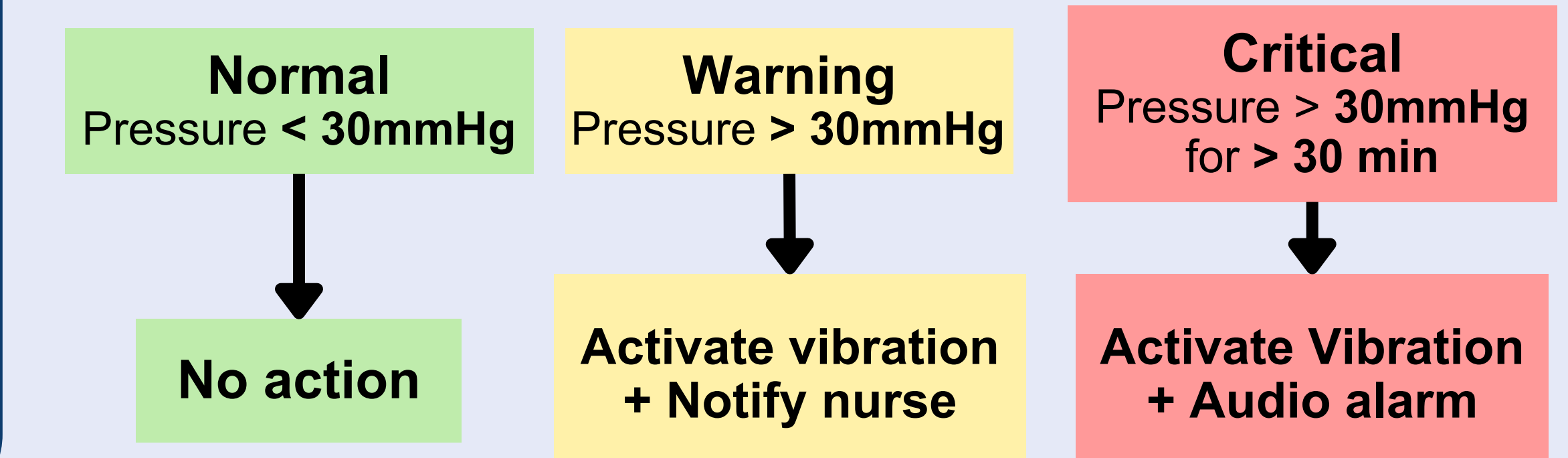
- 2 Connect each patch to control box via wired interface



- 3 GUI with pressure trend for each patch and body map of patch locations



- 4 Vibration is activated to improve capillary flow when pressure exceeds the threshold



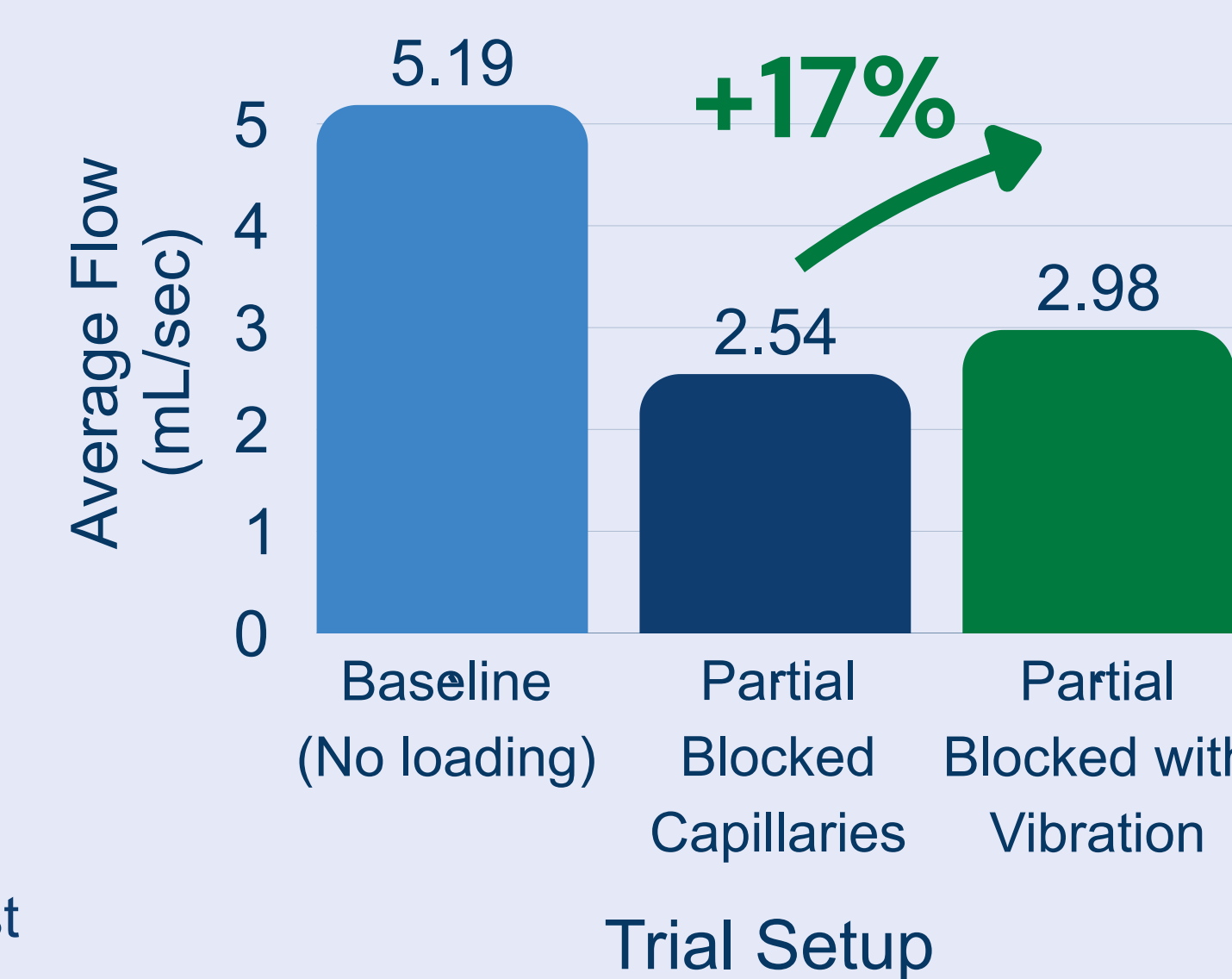
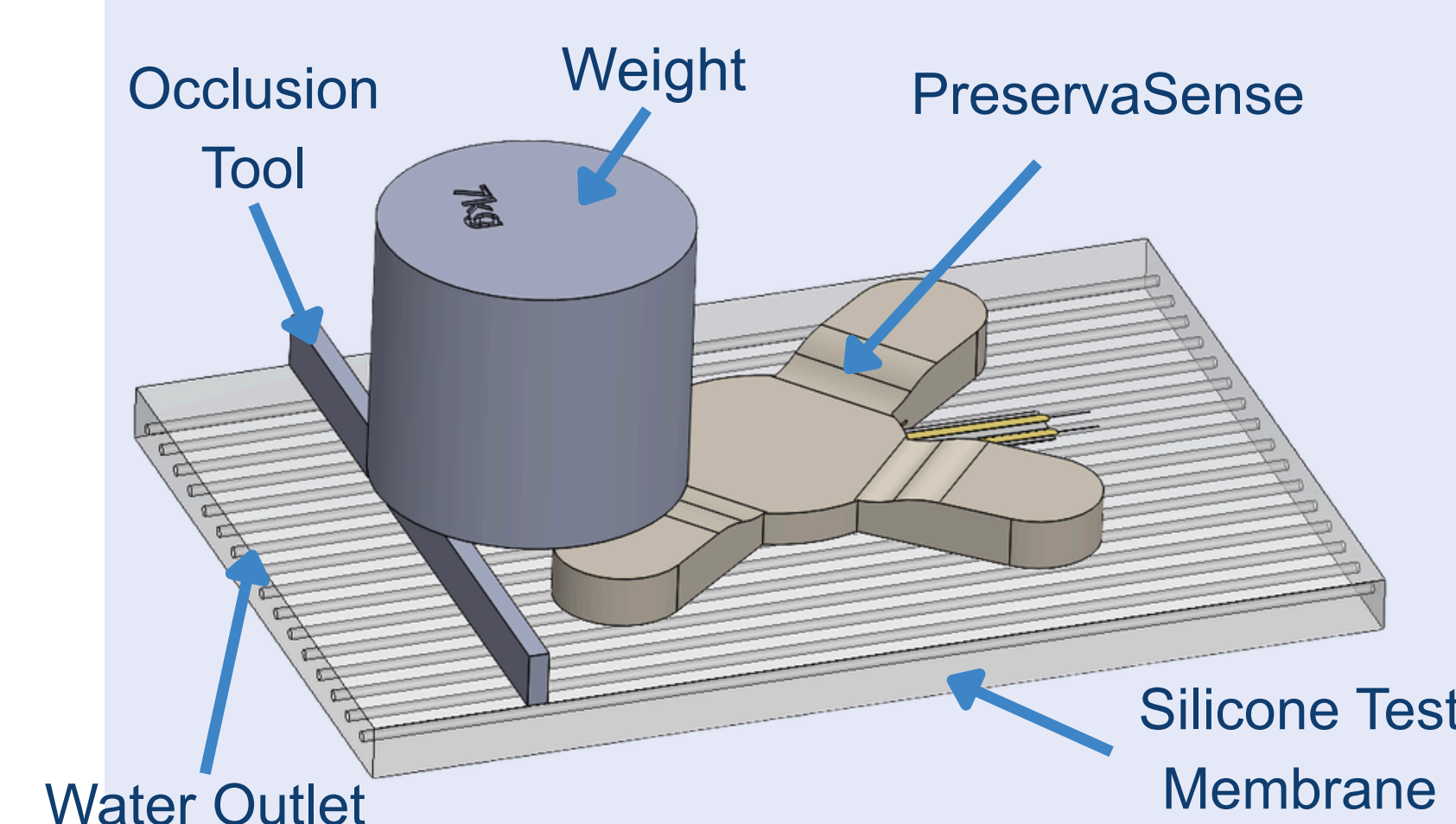
Results

Capillary Flow Rate

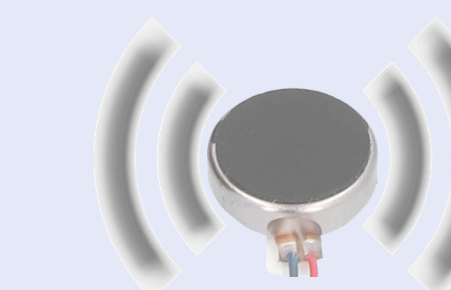
Hypothesis: Vibration induces shear stress along capillaries, promoting turbulent flow → increasing volumetric flow rate

Limitation: In vivo, vibration (~50Hz) reduces blood viscosity + triggers vasodilation → **true flow restoration shown to exceed 80%**^{2,3}

Circulation Test Fixture



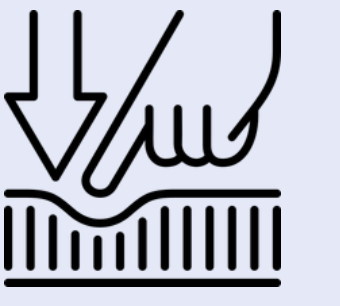
Specification Verification



52.3 Hz
Average Vibration Frequency



0-250 mmHg
Operating Pressure Range



Shore 00-30
Patch Material Hardness

OR Integration Efficacy

4.5/5

Nurse Setup Experience

77%

Adoption if device reduces IPIs up to 40%

4.6/5

Ease of Use

88 sec

Average OR Nurse User Interface Time

100%

Most Likely to use the device if available today

85%

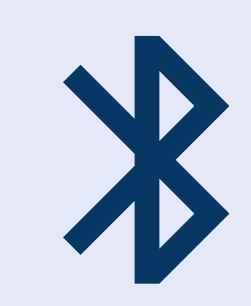
Surgeons find vibration therapy practical

*Interviewed 13 Plastic surgeons & usability testing with 12 OR Nurses from MD Anderson Cancer Center

Next Steps



Refine "warning" and "critical" threshold to ensure complete pressure injury mitigation



Develop a wireless PreservaSense version



Run validation studies to ensure PreservaSense mitigates and monitors IPIs at a % reduction

References

- [1] S. R. H. Zaidi and S. Sharma, "Pressure Ulcer," in StatPearls, Treasure Island (FL)
[2] Ichio S, Yokogawa H, Nakagami G, Sekiya N, Sanada H. In vivo analysis of skin microcirculation and the role of nitric oxide during vibration. PMID: 21918246.
[3] Ren W, et al. Effect of Vibration on Alleviating Foot Pressure-Induced Ischemia under Occlusive Compression. doi: 10.1155/2021/6208499. PMID: 34733455.

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